

CE902 (2025) Worked Solutions — Uploaded Section B

Professional Practice and Research Methodology

This worked-solutions document covers all questions visible in the uploaded PDF: Questions 18 to 21. Section A (Questions 1 to 17) was not present in the uploaded file, so it cannot be solved without additional pages.

Question 18

(a) Likely requirements for the Graduate Software Developer role

Programming and software-development ability. The advert explicitly says the graduate will assist with software applications and integrated systems, so the employer would look for a solid foundation in programming, debugging, version control, and basic software engineering practice.

Ability to work with industrial / integrated systems. Because the company is a systems integrator serving large manufacturers, a strong candidate should be comfortable learning how software interacts with hardware, automation pipelines, data flows, and operational constraints.

Teamwork and willingness to learn. The job stresses learning alongside experienced developers. That suggests the recruiter values coachability, communication, and the ability to work productively in a multi-disciplinary team.

Professional attitude suited to a fast-paced client environment. The advert mentions blue-chip and FTSE100 customers and a fast-paced environment. This means reliability, professionalism, problem solving, and the ability to work accurately under deadlines are all important.

(b) Matching the student statements to the most likely tutor responses

The key idea is that referencing is needed whenever another source has contributed facts, ideas, definitions, wording, or data to the assignment. A bibliography alone is not enough; readers must be able to connect each borrowed point to an in-text citation and then to the reference list.

Statement	Correct response	Reasoning
a	2	Putting a source only in the bibliography does not show where it was used in the assignment. In-text citation is required so the reader can match the source to the relevant claim.
b	4	A wiki is a secondary source. It is usually better to trace the theory back to the original source, but if the wiki is used, it still needs proper referencing.
c	3	Government statistics still require full referencing because readers may need to inspect the source and methodology behind the data.
d	5	Paraphrasing does not remove the need to cite. If another source significantly shaped the ideas used, it must be acknowledged.
e	6	A tutor handout is still somebody else's work and may contain definitions or concepts the reader should be able to trace.
f	1	If no individual author is shown, cite the organisation or other identifiable originator of the source.

Therefore the six pairs are: **a2, b4, c3, d5, e6, f1.**

Question 19

(a) Four general characteristics of a limited company

- It has a **separate legal identity**, meaning the company can own assets, enter contracts, and sue or be sued in its own name.
- Owners normally have **limited liability**, so their personal financial risk is generally limited to the amount invested or guaranteed.
- Ownership is usually represented through **shares or formal capital interests**.
- The company normally has **continuity of existence** independent of changes in shareholders or managers.

(b) Five elements that should appear in a good business plan

- **Executive summary:** a concise overview of the business and its goals.
- **Product/service description:** what the firm will sell and why customers will buy it.
- **Market analysis:** target customers, competitors, demand, and market position.
- **Marketing and sales strategy:** how the firm will win customers and generate revenue.
- **Financial plan:** forecasts, startup costs, cash flow, funding needs, and expected profitability.

(c) Why higher gearing makes a company riskier

Gearing compares the firm's own capital with borrowed funds. When gearing is high, a larger proportion of the business is financed by debt rather than by owners' equity.

Debt creates fixed obligations such as interest payments and principal repayments. Those payments must usually be made even if sales fall, which puts pressure on cash flow.

As a result, a highly geared company is more exposed to downturns, rising interest rates, or delays in customer payments. In weak trading conditions, it can move from reduced profit to financial distress much faster than a low-gearing company.

So the core reason it is considered riskier is that **financial leverage amplifies both gains and losses**, and it increases the chance of liquidity problems or insolvency.

Question 20

(a) Hourly labour cost per employee

First compute the number of productive hours available per employee in one year. A standard working year is taken as 52 weeks $\times$ 5 working days = 260 working days.

Non-working / non-productive days = 8 public holidays + 25 personal holidays + 6 sick days + 12 training days = 51 days

Productive days per employee = 260 - 51 = 209 days

Productive hours per employee = 209 $\times$ 8 = 1672 hours

Annual labour cost per employee = £72,000 / 3 = £24,000

Hourly labour cost = £24,000 / 1672 = £14.35 per hour

Rounded to the nearest pound, the hourly cost for each employee is **about £14 per hour**.

(b) Identify the costing model and compute the final cost

Scenario	Costing model	Calculation	Final cost
(i) Add a flat £4 to a £15 product	Fixed-charge / flat-rate overhead	£15 + £4	£19 per product
(ii) Add 120% overhead to £25 labour	Labour-based percentage overhead	£25 + 1.20 $\times$ £25 = £25 + £30	£55 per hour

(iii) Add 40% overhead to a £60 product cost	Percentage markup / cost-plus overhead	$\text{£}60 + 0.40 \times \text{£}60 = \text{£}60 + \text{£}24$	£84 per product
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Question 21

Present value and net present value analysis

For a single future payment X received after t years at interest rate r , the present value is:

$$PV = X / (1 + r)^t$$

Because £2,000 is paid out now, the net present value is:

$$NPV = \text{PV of repayment} - \text{£}2,000$$

(a) Compare the offer with an alternative return of 6% per year

$$PV \text{ at } 6\% = 2400 / (1.06)^3 = \text{£}2015.09$$

$$NPV \text{ at } 6\% = \text{£}2015.09 - \text{£}2000 = \text{£}15.09$$

The NPV is positive, so at a 6% opportunity cost the promised repayment is worth slightly more than the £2,000 given up now. Therefore the investment is **acceptable**, although the gain is quite small.

(b) Recalculate if an alternative investment yields 10% per year

$$PV \text{ at } 10\% = 2400 / (1.10)^3 = \text{£}1803.16$$

$$NPV \text{ at } 10\% = \text{£}1803.16 - \text{£}2000 = \text{£}-196.84$$

Now the NPV is negative. This means the repayment of £2,400 in three years is worth less, in today's money, than the £2,000 that could instead be invested elsewhere at 10%. So the decision **does change**: at a 10% alternative return, this is **not** a good investment.

End of worked solutions for all questions visible in the uploaded paper.